COS1501 - STUDY UNIT 8

OPERATIONS - COMPLETE TAUGHT NOTES

Binary operations • Properties • Vectors • Matrices • Proofs • Worked examples • Final revision

What this PDF contains
Full lesson-style coverage of Study Unit 8. It includes every official section, the exercise types at the end of the unit, worked examples, extra examples, proof methods, exam methods, common traps, and a final revision checklist.

The unit is about operations: rules that combine mathematical objects. It starts with abstract binary operations, then applies the same ideas to vectors and matrices.

Unit 8 Roadmap

Core idea

A binary operation on a set X takes two elements of X and returns an element of X. Vectors and matrices are then examples of mathematical objects on which useful operations can be defined.

1. Binary Operations

1.1 What is an operation?

An operation is a rule that takes one or more inputs and produces an output. A binary operation takes exactly two inputs. The word binary here means two inputs; it does not mean binary numbers consisting of 0 and 1.

A binary operation on a set X is a function whose domain is X × X and whose codomain is X.

$$* : X \times X \to X$$

If x and y are in X, then x * y must also be in X. This is the closure requirement.

Exam rule
To decide whether a rule is a binary operation on X, ask: (1) are both inputs in X? (2) is the result defined? (3) is the result always in X? One result outside X is enough to fail closure.

Worked Example - A rule that fails closure

Let X={1,2,3} and define x*y=x+y.

1*2=3 is fine because 3 is in X.

But 2*3=5 and 5 is not in X.

*Therefore * is NOT a binary operation on X.*

Extra Example - A rule that is closed

Let X=Z, the integers, and define x*y=x-y.

The difference of two integers is always an integer.

*For all x,y in Z, x-y is in Z. Therefore * is a binary operation on Z.*

1.2 Binary operations as functions

A binary operation is a special kind of function. An ordinary function may have the form f:A->B. A binary operation on X has ordered pairs as inputs, so its domain is X×X, and it returns to X.

For example, union is a binary operation on the power set P(U): two subsets A and B are taken as input, and A union B is another subset of U.

$$f : P(U) \times P(U) \to P(U), \quad f(A,B)=A \text{ union } B$$

1.3 Finite and infinite sets

A finite set has a definite number of elements. For example, A={2,4,6,8} has |A|=4. The empty set is finite because it has 0 elements.

An infinite set has no finite number of elements. Examples include the natural numbers and the real numbers.

Why finite sets matter here
For a finite set, every value of a binary operation can be displayed in an operation table. This makes closure, commutativity and identity elements easier to see.

1.4 Tables for binary operations

In an operation table, the row is the first input and the column is the second input. The entry at their intersection is the result.

*	a	b	c	d
a	a	b	c	d
b	b	c	d	a
c	c	d	a	b
d	d	a	b	c

Example: b*d is found using row b and column d, so b*d=a.

> **Important trap**
> The symbol +, *, triangle, square or any other symbol may simply be the name of the operation. Do not assume it means ordinary arithmetic unless the rule says so.

1.5 List notation for binary operations

A binary operation can also be written as a set of input-output pairs. Because the input itself is an ordered pair, each entry has the form ((x,y), result).

Let X={2,7}. Define an operation # by:

$$\# = \{ ((2,2),2), ((2,7),7), ((7,2),7), ((7,7),2) \}$$

This means 2#2=2, 2#7=7, 7#2=7 and 7#7=2. The same information can be shown in a table:

#	2	7
2	2	7
7	7	2

> **Conversion method**
> List -> table: place each result in row x, column y. Table -> list: read every cell and write ((row heading,column heading),cell value).

Worked Example - Convert a table to list notation

Let X={a,b} and suppose the operation table is: a*a=b, a*b=a, b*a=a, b*b=b.

List every ordered pair in X×X and attach its result.

$$* = \{ ((a,a),b), ((a,b),a), ((b,a),a), ((b,b),b) \}$$

1.6 Unary, binary and n-ary operations

Unary operations take one input. Binary operations take two. Ternary operations take three. More generally, an n-ary operation takes n inputs. Unit 8 concentrates on binary operations.

2. Properties of Binary Operations

Three properties are central in this unit: commutativity, associativity and identity. They are separate properties and must be tested separately.

2.1 Commutative binary operation

$$x*y = y*x \quad \text{for all } x,y \text{ in } X$$

Commutative means changing the order of the two inputs does not change the result.

Ordinary addition and multiplication are commutative. Subtraction is not because 3-5 is not equal to 5-3.

Table shortcut
A finite operation is commutative exactly when its table is symmetric about the main diagonal from top-left to bottom-right.

Worked Example - Checking commutativity from a table

Use the table below. Compare entries mirrored across the main diagonal.

*	1	2	3
1	1	2	3
2	2	3	1
3	3	1	2

1*2=2 and 2*1=2; 1*3=3 and 3*1=3; 2*3=1 and 3*2=1. Every reversed pair agrees, so the operation is commutative.

Worked Example - Disproving commutativity

If one pair gives different answers when reversed, the operation is not commutative.

Example: if a*b=b but b*a=a and a is not b, then a*b is not equal to b*a.

One counterexample is enough to prove NOT commutative.

2.2 Associative binary operation

$$(x*y)*z = x*(y*z) \quad \text{for all } x,y,z \text{ in } X$$

Associative means changing the grouping does not change the result. The order x,y,z stays the same; only the brackets move.

Worked Example - Why subtraction is not associative

(10-5)-2 = 5-2 = 3.

10-(5-2) = 10-3 = 7.

3 is not equal to 7, so subtraction is not associative.

Do not mix them up
Commutative = order changes. Associative = grouping changes. An operation can have either property, both properties, or neither.

For a finite set X with n elements, a complete brute-force associativity check has n^3 ordered triples. If X={1,2}, there are 2^3=8 triples.

Complete 2-element associativity method

Suppose the table is:

*	1	2
1	1	2
2	2	1

Check all triples (1,1,1), (1,1,2), (1,2,1), (1,2,2), (2,1,1), (2,1,2), (2,2,1), (2,2,2). For each, calculate both (x*y)*z and x*(y*z). If all eight match, the operation is associative.

Extra Example - One associativity check from the table

For x=1,y=2,z=2:

$$(1*2)*2 = 2*2 = 1$$

$$1*(2*2) = 1*1 = 1$$

The two sides agree for this triple. A full proof for the 2-element set checks all eight triples.

2.3 Identity element

$$e*x = x*e = x \quad \text{for all } x \text{ in } X$$

An identity element leaves every element unchanged, from both the left and the right.

For ordinary addition, 0 is the identity. For ordinary multiplication, 1 is the identity.

Table shortcut
For e to be an identity, row e must reproduce the column headings in order, and column e must reproduce the row headings in order. Both are required.

Worked Example - Finding an identity in a table

Use the commutative 3-element table shown earlier.

Row 1 is 1,2,3 and column 1 is 1,2,3.

Therefore 1*x=x and x*1=x for every x in the set.

$$1 \text{ is the identity element.}$$

Identity elements are unique

Suppose e and f are both identities. Because e is an identity, e*f=f. Because f is an identity, e*f=e. Therefore e=f. So there cannot be two different identity elements for the same operation.

2.4 Property comparison

Property	Test	Meaning
Commutative	x*y = y*x	Changes order
Associative	(x*y)*z = x*(y*z)	Changes grouping
Identity	e*x = x*e = x	Leaves every element unchanged

3. Operations on Vectors

3.1 What is a vector?

In this course, a vector is an ordered n-tuple of numbers, with n at least 2.

$$u = (u1,u2,...,un)$$

Order matters: (2,5) is not the same vector as (5,2). The number of coordinates is the dimension of the vector.

3.2 Vector sum

Two vectors can be added only when they have the same number of coordinates. Add corresponding coordinates.

$$u+v = (u1+v1, u2+v2, ..., un+vn)$$

Worked Example - Vector addition

Let u=(1,2,3) and v=(4,5,6).

$$u+v = (1+4,2+5,3+6) = (5,7,9)$$

Extra Example - Vector subtraction

Let u=(7,5,3) and v=(2,1,4).

$$u-v = (7-2,5-1,3-4) = (5,4,-1)$$

Undefined case

(1,2,3)+(4,5) is undefined because the vectors have different dimensions.

3.3 Scalar-vector product

A scalar is an ordinary number. To multiply a vector by a scalar, multiply every coordinate by the scalar.

$$r\,u = (r\,u1, r\,u2, ..., r\,un)$$

Worked Example - Scalar-vector multiplication

Let u=(2,-3,5) and r=-2.

$$-2u = (-4,6,-10)$$

Worked Example - Combining vector operations

Let u=(2,-1,4) and v=(3,5,-2). Find 3u-2v.

$$3u=(6,-3,12)$$

$$2v=(6,10,-4)$$

$$3u-2v=(0,-13,16)$$

3.4 Dot product / inner product

The dot product takes two vectors of the same dimension and returns a scalar. Multiply corresponding coordinates and then add those products.

$$u \cdot v = u1v1 + u2v2 + ... + unvn$$

Worked Example - Dot product

Let u=(2,4,6) and v=(1,3,5).

$$u \cdot v = 2(1)+4(3)+6(5) = 2+12+30 = 44$$

Extra Example - Dot product with negative values

Let u=(2,-1,4) and v=(3,5,-2).

$$u \cdot v = 2(3)+(-1)(5)+4(-2) = 6-5-8 = -7$$

Key distinction
Vector addition gives a vector. Scalar-vector multiplication gives a vector. The dot product of two vectors gives a scalar.

Extra Example - Another combined vector exercise

Let u=(3,1), v=(-4,-4) and w=(0,-1). Find -3(v+w).

$$v+w=(-4,-5)$$

$$-3(v+w)=(12,15)$$

Application idea

If one vector stores quantities and another stores corresponding prices, their dot product gives the total cost. This shows why the dot product is useful in computing and data representations.

4. Operations on Matrices

4.1 What is a matrix?

A matrix is a rectangular array of numbers arranged in rows and columns. A matrix with m rows and n columns has size m×n.

$$A = [a\_ij], \quad where\ i = row\ number\ and\ j = column\ number$$

For a_23, the first index 2 means row 2 and the second index 3 means column 3.

The positions of matrix entries matter: changing an entry from one row or column to another changes the information represented by the matrix. A matrix with one row is called a row matrix, and a matrix with one column is called a column matrix.

Worked Example - Dimensions and entries

For A=[[1,2,3],[4,5,6]], A has 2 rows and 3 columns, so A is 2×3.

The entry a_23 is row 2, column 3, so a_23=6.

4.2 Matrix addition

Two matrices can be added only when they have exactly the same dimensions. Add corresponding entries.

$$A+B = [a\_ij + b\_ij]$$

Worked Example - Matrix addition

Let A=[[1,2],[3,4]] and B=[[5,6],[7,8]].

$$A+B = [[6,8],[10,12]]$$

Undefined case
A 2×2 matrix cannot be added to a 2×3 matrix. Same dimensions are required.

4.3 Scalar-matrix multiplication

Multiply every entry of the matrix by the scalar.

$$rA = [r\ a\_ij]$$

Worked Example - Scalar-matrix multiplication

Let A=[[2,-1],[3,4]].

$$3A = [[6,-3],[9,12]]$$

4.4 Matrix multiplication - the dimension rule

Matrix multiplication is not entry-by-entry. It uses rows of the first matrix and columns of the second matrix.

If A is m×n and B is p×q, then AB exists only when n=p. If it exists, the result is m×q.

$$(m\times n)(n\times p)\ \text{->}\ m\times p$$

Fast exam method
Write the dimensions next to each factor before calculating. The two middle dimensions must match. The two outside dimensions give the size of the answer.

Worked Example - Dimension check

A is 2×3 and B is 3×4.

$$(2\times 3)(3\times 4)\ \text{->}\ 2\times 4$$

The middle 3s match, so AB exists and is 2×4.

Extra Example - Invalid multiplication

A is 2×3 and B is 2×4.

The middle dimensions 3 and 2 do not match.

$$AB \text{ is undefined.}$$

4.5 How matrix multiplication works

Each entry of AB is the dot product of one row of A with one column of B.

Worked Example - Full 2×2 multiplication

Let A=[[1,2],[3,4]] and B=[[5,6],[7,8]].

Top-left: 1(5)+2(7)=19.

Top-right: 1(6)+2(8)=22.

Bottom-left: 3(5)+4(7)=43.

Bottom-right: 3(6)+4(8)=50.

$$AB = [[19,22],[43,50]]$$

Worked Example - 2×3 multiplied by 3×2

Let A=[[1,2,3],[4,5,6]] and B=[[7,8],[9,10],[11,12]].

Entry (1,1): 1(7)+2(9)+3(11)=58.

Entry (1,2): 1(8)+2(10)+3(12)=64.

Entry (2,1): 4(7)+5(9)+6(11)=139.

Entry (2,2): 4(8)+5(10)+6(12)=154.

$$AB = [[58,64],[139,154]]$$

4.6 Matrix multiplication is not generally commutative

For ordinary numbers, ab=ba. Matrix multiplication does not generally have this property.

Worked Example - AB is not equal to BA

Let A=[[1,2],[3,4]] and B=[[4,3],[2,1]].

$$AB = [[8,5],[20,13]]$$

$$BA = [[13,20],[5,8]]$$

Since the results differ, matrix multiplication is not commutative.

Even stronger warning
AB may exist while BA does not. Reversing factors changes the dimension test.

4.7 Zero matrix

The zero matrix has 0 in every entry. It is the identity element for matrix addition: A+O=O+A=A.

$$O\_2 = [[0,0],[0,0]]$$

Also, multiplying a zero matrix by any appropriately sized matrix gives a zero matrix. For example, every row-column dot product contains only zeros, so every resulting entry is zero.

$$OA=O \text{ and } AO=O \quad \text{whenever the products are dimensionally defined}$$

4.8 Identity matrix

The identity matrix I_n is an n×n matrix with 1s on the main diagonal and 0s elsewhere. It is the identity for matrix multiplication.

$$I\_2 = [[1,0],[0,1]], \quad AI\_2 = I\_2A = A$$

5. Important Unit 8 Exercise Types and Formal Proofs

5.1 Construct X and Y so XY is 3×3 but YX is 2×2

This is a pure dimension-construction problem. We want the outside dimensions of XY to be 3 and 3, and the outside dimensions of YX to be 2 and 2.

Choose X to be 3×2 and Y to be 2×3.

$$XY\text{: } (3\times 2)(2\times 3) \text{ -> } 3\times 3$$

$$YX\text{: } (2\times 3)(3\times 2) \text{ -> } 2\times 2$$

For example:

$$X=[[1,0],[0,1],[1,1]], \quad Y=[[1,2,3],[4,5,6]]$$

The actual entries can vary. What matters first is the dimensions.

Exam insight
When a question asks you to PROVIDE matrices with particular product sizes, solve the dimension puzzle before choosing entries. Simple 0s and 1s are usually easiest.

5.2 Non-zero matrices whose product is the zero matrix

Unlike ordinary real numbers, non-zero matrices can multiply to give the zero matrix. This is an important warning: matrices can have zero divisors.

Choose:

$$X=[[1,1],[1,1]], \quad Y=[[1,-1],[-1,1]]$$

Neither X nor Y is the zero matrix, but:

$$XY = [[1(1)+1(-1), 1(-1)+1(1)], [1(1)+1(-1), 1(-1)+1(1)]]$$

$$XY = [[0,0],[0,0]]$$

Meaning
You cannot blindly use the ordinary-number rule "xy=0 implies x=0 or y=0" for matrices. That rule fails for matrix multiplication.

5.3 Formal proof: matrix addition is commutative

Let A and B be arbitrary 2×2 matrices:

$$A=[[a,b],[c,d]], \quad B=[[e,f],[g,h]]$$

Then:

$$A+B = [[a+e,b+f],[c+g,d+h]]$$

Real-number addition is commutative, so a+e=e+a, b+f=f+b, c+g=g+c and d+h=h+d. Therefore:

$$A+B = [[e+a,f+b],[g+c,h+d]] = B+A$$

Therefore matrix addition is commutative on the set of 2×2 matrices.

5.4 Formal proof: additive identity for 2×2 matrices

Let:

$$O=[[0,0],[0,0]]$$

For arbitrary A=[[a,b],[c,d]]:

$$A+O = [[a+0,b+0],[c+0,d+0]] = [[a,b],[c,d]] = A$$

$$O+A = A$$

Therefore O is the identity element for addition on the set of 2×2 matrices.

5.5 Formal proof: matrix multiplication is not commutative

To prove a property is NOT universal, one counterexample is enough. Use:

$$A=[[1,2],[3,4]], \quad B=[[4,3],[2,1]]$$

$$AB=[[8,5],[20,13]], \quad BA=[[13,20],[5,8]]$$

$$AB \text{ is not equal to } BA.$$

Therefore multiplication is not a commutative operation on the set of 2×2 matrices.

5.6 Formal proof: multiplicative identity for 2×2 matrices

Let:

$$I=[[1,0],[0,1]]$$

For arbitrary A=[[a,b],[c,d]]:

$$AI = [[a(1)+b(0),a(0)+b(1)],[c(1)+d(0),c(0)+d(1)]] = [[a,b],[c,d]] = A$$

Similarly:

$$IA = A$$

Therefore I is the identity element for matrix multiplication on the set of 2×2 matrices.

6. Consolidated Worked Examples

6.1 Binary operation: closure

Question: Let X={0,1,2} and define x*y=(x+y) mod 3. Is this a binary operation on X?

Possible remainders modulo 3 are only 0,1,2. Therefore every result lies in X. So the operation is closed and is a binary operation on X.

6.2 Reading an operation table

*	a	b	c
a	a	b	c
b	c	a	b
c	b	c	a

b*c is row b, column c, so b*c=b. c*b is row c, column b, so c*b=c. Since b is not c, this table is not commutative.

6.3 Identity from a table

*	a	b	c
a	a	b	c
b	b	c	a
c	c	a	b

Row a reproduces a,b,c and column a also reproduces a,b,c. Therefore a is the identity.

6.4 Vector calculation

Let u=(4,-2,1) and v=(-1,3,5). Find 2u+3v and u·v.

$$2u=(8,-4,2),\quad 3v=(-3,9,15)$$

$$2u+3v=(5,5,17)$$

$$u\cdot v=4(-1)+(-2)(3)+1(5)=-4-6+5=-5$$

6.5 Matrix dimension reasoning

If A is 4×2 and B is 2×5, then AB exists and is 4×5. BA would require (2×5)(4×2), but 5 and 4 do not match, so BA does not exist.

6.6 Mixed matrix operations

Let A=[[2,-1],[0,3]] and B=[[1,4],[5,-2]].

$$A+B=[[3,3],[5,1]]$$

$$2A=[[4,-2],[0,6]]$$

$$AB=[[2(1)+(-1)(5),\ 2(4)+(-1)(-2)],\ [0(1)+3(5),\ 0(4)+3(-2)]]$$

$$AB=[[-3,10],[15,-6]]$$

7. Common Exam Traps

- Binary means two inputs, not base-2 numbers.
- Closure must hold for every possible pair. One output outside X means the rule is not a binary operation on X.
- In an operation table, use row first and column second.
- The operation symbol does not automatically mean ordinary addition or multiplication.
- One matching pair does not prove commutativity; all pairs must match. One mismatch disproves it.
- Commutativity changes order; associativity changes grouping.
- An identity must work from both sides.
- Vector addition and dot products require matching dimensions.
- A dot product returns a scalar, not a vector.
- Matrix addition requires identical dimensions.
- Matrix multiplication does not require square matrices.
- For AB, columns of A must equal rows of B.
- AB existing does not imply BA exists.
- Matrix multiplication is row-by-column, not entry-by-entry.
- Matrix multiplication is not generally commutative.
- Non-zero matrices can multiply to the zero matrix.
- The zero matrix is the additive identity; the identity matrix is the multiplicative identity.

8. Master Formula and Method Sheet

Binary operations

$$* : X \times X \to X$$

$$\text{Commutative: } x*y = y*x$$

$$\text{Associative: } (x*y)*z = x*(y*z)$$

$$\text{Identity: } e*x = x*e = x$$

Vectors

$$u+v=(u1+v1,...,un+vn)$$

$$r\,u=(r\,u1,...,r\,un)$$

$$u\cdot v=u1v1+...+unvn$$

Matrices

Size m×n means m rows and n columns

$$A+B=[a\_ij+b\_ij] \quad \text{(same size required)}$$

$$rA=[r\,a\_ij]$$

$$(m\times n)(n\times p) \to m\times p$$

$$A+O=A$$

$$AI=IA=A$$

Fast answer procedures

If asked: Is this a binary operation?

1. Identify X.
2. Check every input pair belongs to X×X.
3. Check every output belongs to X.
4. If one output leaves X, say not closed and therefore not a binary operation on X.

If asked: Is the table commutative?

5. Compare entries reflected across the main diagonal.
6. If all match, it is commutative.
7. If one mismatch appears, give it as a counterexample.

If asked: Is it associative?

8. Write (x*y)*z and x*(y*z).
9. For a small finite set, check all triples unless a shortcut/proof is available.
10. A single failing triple disproves associativity.

If asked: Find the identity

11. Find a row that reproduces the headings.
12. Check the same element also has a column reproducing the headings.
13. State e*x=x*e=x.

If asked: Can matrices multiply?

14. Write the dimensions: (m×n)(p×q).
15. Check n=p.
16. If yes, result is m×q.
17. Then use row-by-column dot products.

9. Final Revision Section

9.1 What you must be able to explain in words

- What a binary operation is and why closure matters.
- The difference between finite and infinite sets.
- How list notation and operation tables represent the same binary operation.
- The difference between commutative, associative and identity properties.
- What a vector and a scalar are.
- How vector addition, scalar-vector multiplication and dot products work.
- What a matrix is, how dimensions are read, and what a_{ij} means.
- How matrix addition, scalar multiplication and matrix multiplication work.
- Why matrix multiplication is not generally commutative.
- Why non-zero matrices may multiply to the zero matrix.
- The roles of the zero matrix and the identity matrix.

9.2 Mini self-test

1. Let X={1,2,3} and x*y=x+y. Is * a binary operation on X? Why?

2. Convert #={((a,a),b),((a,b),a),((b,a),a),((b,b),b)} into a table.

3. State the definition of a commutative binary operation.

4. State the definition of an associative binary operation.

5. What two table checks are needed for an identity element?

6. Calculate (2,-1,4)+(3,5,-2).

7. Calculate 3(2,-1,4)-2(3,5,-2).

8. Calculate (2,-1,4)·(3,5,-2).

9. Can a 2×3 matrix be added to a 2×2 matrix?

10. What size is (4×3)(3×2)?

11. Why can AB exist while BA does not?

12. Give dimensions for X and Y so XY is 3×3 and YX is 2×2.

13. Can non-zero matrices multiply to the zero matrix?

14. State the 2×2 additive identity matrix.

15. State the 2×2 multiplicative identity matrix.

9.3 Answers

1. No. For example 2*3=5, and 5 is not in X, so closure fails.

2. Rows/columns a,b with entries: row a -> b,a; row b -> a,b.

3. x*y=y*x for every x,y in X.

4. (x*y)*z=x*(y*z) for every x,y,z in X.

5. Row e must reproduce the headings AND column e must reproduce the headings.

6. (5,4,2).

7. (0,-13,16).

8. -7.

9. No. Matrix addition requires exactly the same dimensions.

10. 4×2.

11. Reversing the matrices changes which dimensions must match.

12. X:3×2 and Y:2×3.

13. Yes. Matrices can have zero divisors.

14. [[0,0],[0,0]].

15. [[1,0],[0,1]].

9.4 Completion checklist

- ✓ Define a binary operation and closure.
- ✓ Distinguish finite and infinite sets.
- ✓ Read an operation table correctly.
- ✓ Convert between list notation and table notation.
- ✓ Test commutativity.
- ✓ Test associativity.
- ✓ Find an identity element.
- ✓ Add and subtract vectors.
- ✓ Multiply a vector by a scalar.
- ✓ Calculate a dot product.
- ✓ Recognise invalid vector operations.
- ✓ Identify matrix dimensions and entries.
- ✓ Add matrices.
- ✓ Multiply a matrix by a scalar.
- ✓ Check matrix-multiplication compatibility.
- ✓ Calculate matrix products row-by-column.
- ✓ Explain why AB is generally not BA.
- ✓ Construct matrices from required product dimensions.
- ✓ Give an example of non-zero matrices with zero product.
- ✓ Prove matrix addition is commutative.
- ✓ Prove the zero matrix is the additive identity.
- ✓ Disprove commutativity of matrix multiplication by counterexample.
- ✓ Prove the identity matrix is the multiplicative identity.

Final verdict

If you can do every item in this checklist without looking at the notes, you have covered the full content and exercise types of COS1501 Study Unit 8.

Source basis

These notes were compiled from the COS1501 Study Unit 8 structure and exercises, with expanded explanations and additional worked examples for study and exam preparation. They are teaching notes, not a verbatim reproduction of the source material.